Yiwen Chen

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EDUCATION

- **M.Sc., China University of Geosciences (Wuhan)** 09/2022-06/2025

Computer Science **GPA: 92.03125/100 Rank(3/191)**

- **B.S., Beijing Language and Culture University** 09/2018-06/2022

Digital Media Technology **GPA:3.62/4.0 Top 10%**

PUBLICATIONS

Article: Yi-wen Chen, Si-qi Hu, Ming Li, Xue-song Yan, Graph convolutional network for water network partitioning, *Expert Systems with Applications*, Volume 261,2025,125502, ISSN 0957-4174.
As First Author, Accepted, (JCR Q1, IF=7.5)
DOI: https://doi.org/10.1016/j.eswa.2024.125502

Patent: A method for optimizing the layout of sensors in a graph neural network water supply network
As First Author, CN202311667909.8

RESEARCH EXPERIENCE

Summer Research, advised by Prof. Yuke Wang, Rice university 07/2023-08/2023

- Read articles that focus on how to improve the speed and performance of large language model inference such as Gemini, DataStates-LLM, ServerlessLLM, PagedAttention
- Learn GPU programming and CUDA, reproduce mamba-minimal

Graph Convolutional Network for Water Network Partitioning 09/2023-06/2024

- Leader of this project, in charge of idea formulation, implementation, experiments, analysis and writing
- By reading a large amount of literature, I found it is essential to utilize the GCN algorithm to combine pressure data and structural topology information comprehensively to cluster the network nodes which nearly nobody has done before.
- Learning materials about deep learning and hydrology, I simulated and generated datasets with Python code.
- On the PyTorch platform, I tried and conducted various experiments about different GCN methods such as Structural Deep Clustering Networks and KMeans on different water networks.
- By analyzing the result, I found a Cross-Attention Fusion Based Enhanced Graph Convolutional Network for Clustering performance well, thus I chose it as my method.
- Write the manuscript by Latex and draw pictures with Visio and Python.

A method for optimizing the layout of sensors in a graph neural network water supply network 09/2023-01/2024

- Use graph convolutional network clustering algorithm to partition the water supply network into sub-regions, and arrange sensors in each region after partitioning, in order to better detect the entire water supply network system.

The 6th/7th Academic Conference on Intelligent Optimization and Scheduling ***Wall Exhibition*** 2024/2023

- Write a review, titled "A review of leak detection and localization methods for complex water supply systems"
- Classify and deeply analyze the literature from the perspectives of physical sensors, data-driven and model-driven for the current important research results in this field, and discuss the directions that can be studied in the future

The 15th Guangdong Hong Kong Macao Greater Bay Area Doctoral Student Nanshan Academic Forum and Tsinghua University 706th Doctoral Student Academic Forum ***Wall Exhibition*** 06/2023

Taobao Double Eleven Big Data Analysis and Prediction 04/2023-06/2023

- Various typical operations involved in the entire data processing process of data preprocessing, storage, querying, and visualization analysis of 50 million Taobao data in the Linux system

WeChat Personalized Signature Mining and Word Cloud Visualization 01/2021-02/2021

- Use python, wordcloud, jieba, ichat to visualize my 1000+ wechat friends' personalized signature in a custom mask

AI Speech Recognition Research 04/2020-04/2021

Leader of College Students' Innovative Entrepreneurial Training Program, Beijing Municipal Level

- Building and training a Chinese tone recognition model and published a simple speech recognition mini program.
- On the PyTorch platform, we extracted tone information features from the Tsinghua 30h dataset, and established a CNN+CTC model through VScode's remote SSH to connect remote service to use GPU for tone recognition training.
- Developed a speech recognition mini Wechat program by learning open-source code.
- Gained knowledge in areas such as Speech Signal Processing, Fourier Transform, Neural Networks, and Mini program development
- Write a final thesis about this project

English Learning System Based on Virtual Reality *Undergraduate graduation project* 11/2021-04/2022

- Use C#, SteamVr, Unity, VRTK and Vive Pro to build an English learning system
- I realize the functions of word collection and statistics, character interaction and dialogue, etc
- Users can use handles to interact with the objects and NPCs in the virtual world
  Can be visited by https://youtu.be/-ZioAZY1JSs

ADDITIONAL EXPERIENCE

University of California Berkeley Summer Visit 07/2023-08/2023

- Learning Data Structures and Algorithms CS61b Course in EECS and Marketing class in Haas
- Know how to use java to produce game 2048 and use java to realize a simple version of Git

Summer School of Peking University 07/2022-08/2022

- Learning Elective Course on Intelligent Computing Systems, scored 93

Volunteer for the 3rd Youth Academic Seminar of the Cross Science Department of the *National Natural Science Foundation of China* 10/2023

Head of the Academic Department of the Graduate Association of the School of Computer Science 2022

HONORS & AWARDS

- A member of Baidu AI Pinecone Talent Development Elite Class 2024
- Successfully passed the WPS office C++ back-end school-enterprise cooperation training camp 2024
- Academic Scholarship (First Class) for three consecutive years 2022-2024
- High School Information Technology Teaching Qualification Certificate 2023
- Speaker on behalf of the incoming graduate student 2022
- Outstanding youth league cadres at the school level 2022/2018
- S Award in the American Mathematical Modeling Competition for College Students 2021
- Merit student of Beijing language and culture university 2021

PROFESSIONAL SKILLS

Good at Mathematics:

- Second Prize in the Beijing Division of the 12th National College Student Mathematics Competition 2020
- Second prize in the Hubei Division of the National Mathematics/ Chemistry / Biology Olympiad 2017

Programming: C/C++/Python/JAVA/C #/Matlab/CUDA, master Object-Oriented programming concepts.

Tools: Git/CMake/Gdb/Vim/Springboot/Qt/Unity/3D-Max/Spark/Hadoop

Data Analysis&Visualization: CSS/Javascript/Vue/Html/Echarts/SPSS

Good English Ability: IELTS 7/ Test for English Major-4

PERSONAL SUMMARY

- I enjoy technical communication and often browse and learn on websites such as Blogs, Zhihu, Github, Bilibili.
- Passionate about learning, honest and upright, with the ability to conduct independent or cooperative research.